

SHIVASAI KRISHNA GOUD BATHINI

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Professional Summary

AI /ML Engineer & Applied Reasearcher — Building Production-Scale ML Systems

An Engineer focused on making complex problems accessible through better systems. Specializes in building scalable ML infrastructure for time-series forecasting and RAG applications, bridging the gap between research innovation and practical deployment.

Technologies

Languages & Development : Python | Java | C/C++ | Git | CI/CD | Docker | Kubernetes | MLflow

Machine Learning & AI: Deep Learning | Neural Networks (CNNs, Transformers, LLMs) | GenAI | LLMs | NLP | Computer Vision | RAG | Statistical Modeling | Time-Series Forecasting

Frameworks & Cloud Platforms: PyTorch | TensorFlow | Databricks | scikit-learn | Hugging Face Transformers | LangChain | FastAPI | Flask | Pandas | NumPy | Redis | AWS (S3, EC2, Lambda, SageMaker, Bedrock) | Azure AI ML | OpenAI | Kubeflow | MLOps | VectorDB | LlamaIndex | VertexAI | Kafka | MCP servers | Azure Foundry

Relational and NoSQL databases: PostgreSQL | MongoDB | ElasticSearch

Experience

Applied Research Engineer, University of Bridgeport – Bridgeport, CT

Sep 2023 – Dec 2025

- Designed and deployed a domain-specific Large Language Model (LLM) supporting academic, administrative, and student-facing services across the University's internal ecosystem, serving 10K+ daily requests
- Implemented role-based access control (RBAC) using Azure Active Directory authentication, automatically identifying faculty and student accounts via institutional email domains and granting role-appropriate access, with elevated privileges for faculty users.
- Built and maintained an end-to-end **MLOps and CI/CD pipeline** using **Kubeflow and Kubernetes**, automating data ingestion, model training, validation, deployment, and retraining triggers on an on-prem **HPC cluster (8× NVIDIA H200 GPUs) and Azure ML based workflows**.
- Enabled real-time model observability by integrating inference logging with Microsoft Azure monitoring and visualization tools, providing live graphs for usage, latency, and performance to track model health in production.
- Designed **GPU-aware workload orchestration and priority-based scheduling** with **Kubeflow Pipelines**, balancing interactive inference workloads with background training and evaluation jobs to optimize GPU utilization.
- Integrated Microsoft Graph APIs with SharePoint and Teams content to enforce secure access patterns and enable identity-aware retrieval in AI-powered applications.
- Built data engineering pipelines using Apache Spark (Databricks) for large-scale data preprocessing, transformation, and dataset versioning feeding downstream LLM training and inference pipelines

Software Infrastructure Engineer – Platform & GTM Systems, Vojoy Games Pvt. Ltd. –
Hyderabad, India (Hybrid)

May 2021 – Jul 2023

- Built and maintained cloud-based platform services on AWS and Azure to support game backend systems, analytics workloads, and go-to-market (GTM) experimentation across multiple mobile game releases.
- Designed CI/CD pipelines using Azure DevOps and GitHub Actions to automate build, test, and deployment workflows for game services, analytics jobs, and internal tooling, improving release reliability and reducing manual deployment overhead.
- Developed data pipelines and backend services to collect, clean, and aggregate gameplay and marketing data, enabling statistical analysis of user acquisition, retention, and ad performance.
- Implemented forecasting and trend analysis models using classical statistical techniques (ARIMA, SARIMA, linear regression) to support download growth projections, ad revenue estimation, and regional campaign planning.
- Built experimentation and reporting workflows to support A/B testing of gameplay features and ad strategies, surfacing results to product and GTM teams through dashboards and scheduled reports. .
- Introduced AI-assisted tooling in late 2022 by integrating lightweight NLP models and external APIs for tasks such as log classification, anomaly flagging, and content QA support, operating primarily as internal productivity tools rather than customer-facing features.
- Optimized infrastructure costs through environment standardization, autoscaling policies, and resource right-sizing, balancing performance needs with operational efficiency.
- Collaborated closely with product, analytics, and GTM teams to translate business questions into reliable platform capabilities, enabling data-driven decisions without overengineering solutions.

Projects/Applied Experience

Stock Forecasting System — Mixture-Based Time Series Modeling with LLM-Guided Agents (M.S. Research Project)

Sep 2025 – Present

- Developed a confluent MoE architecture, where the gating network itself is transformer-based. Along with this, I incorporated linear, transformer, xLSTM, and minGRU-based approaches, which we adapt for time-series forecasting (TSF) within the MoE framework.
- Conducted empirical evaluations on various datasets to demonstrate the effectiveness of our MoE approach for TSF.

Personal Attorney — Agentic Retrieval-Augmented Legal Reasoning Framework (Research Project)

Jan 2025 – Aug 2025

- Architected an AI-powered legal reasoning system using Azure OpenAI-based Retrieval-Augmented Generation (RAG), enabling grounded legal analysis over large-scale statutory, case law, and contract documents.
- Embedded and indexed 50K+ legal documents using Azure AI Search with vector and hybrid search, enabling high-precision semantic retrieval beyond traditional keyword-based methods.
- Applied Non-negative Matrix Factorization (NMF) to uncover 30+ latent legal topics, supporting large-scale case clustering and legal trend discovery.(.....**READ MORE**)

Custom Sanskrit Transformer — High-Performance Language Modeling System

Apr 2024 – Nov 2024

- Tuned a Sanskrit dataset using a custom 25,000-token vocabulary with Byte Pair Encoding (BPE), improving token efficiency and semantic representation for complex compound words.
- Integrated TensorRT, CUDA, and cuDNN with FP16/INT8 mixed precision, achieving $2\times$ training/inference speedup and 35% GPU memory reduction.
- Optimized sequential batching and attention masks for fine-tuning and semantic analogy prediction across 1M+ token-level samples.

Cognitive HOI Reasoning Research – Agentic AI System

Apr 2022 – Mar 2023

- Built a compositional 3D perception framework inspired by EVA3D NeRF using Bongard-HOI and HAKE datasets, emphasizing human-level cognitive reasoning over recognition.
- Generated 100+ few-shot instances from structured human-object visual relationships to enable compositional concept learning under limited supervision.
- Evaluated 5+ state-of-the-art few-shot methods (non-episodic and meta-learning) across 10+ reasoning benchmarks, demonstrating robust generalization.